

# [DC]Toward Adaptive and Accessible XR Through User and Context Modeling

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## ABSTRACT

As XR technologies become increasingly integrated into everyday applications, designing interfaces that dynamically adapt to diverse users, tasks, and environments is critical. Traditional XR systems often assume static layouts and uniform user capabilities, but in real-world use, users move between contexts, switch tasks, and face varying physical and cognitive demands, which can disrupt usability. My research investigates adaptive interaction in XR, focusing on dynamically adjusting spatial layouts, input strategies, feedback mechanisms, and interaction logic to respond to users' physical, cognitive, and environmental conditions. My PhD research examines age-related differences, develop hand redirection techniques to optimize interaction in supine postures in VR, and explore adaptive mixed-reality learning environments that tune task complexity and visual feedback based on cognitive load.

**Index Terms:** Extended Reality, Adaptive Interface.

## 1 INTRODUCTION

As XR technologies become increasingly integrated into everyday applications, designing user interfaces that dynamically adapt to diverse users, tasks, and environments is a growing research priority. Traditional XR systems often assume static interaction contexts, fixed layouts, and uniform user capabilities [9, 8, 4]. However, in real-world use, these assumptions break down: users move between environments, switch tasks rapidly, and interact under varying physical and cognitive conditions. For example, a desk worker using a headset might spread virtual windows across their workspace to monitor emails, documents, and chat applications. When a teammate unexpectedly stops by for an in-person discussion, the worker suddenly needs space to see both the physical colleague and only the most relevant digital content. Without adaptation, virtual apps may clutter the view, block eye contact, or remain anchored in inconvenient positions. To maintain usability and relevance, XR interfaces must be capable of adapting in real-time to such variability.

There is increasing interest in making these systems more adaptive to users and their environments. Adaptation in user interfaces has a long history, particularly within 2D graphical user interfaces, where numerous approaches have been proposed to optimize layouts and interaction through algorithmic methods and interface adaptation strategies [2, 5]. Unlike 2D interfaces, adaptive systems in immersive environments are more complicated, since they involve dynamically adjusting UI, interaction techniques, and content placement based on factors such as user context (e.g., physical position, cognitive load, task) and environmental conditions (e.g., lighting, spatial constraints) [3]. Adaptive interfaces in XR can adjust content, layout, and interaction based on context, user behavior, or task demands, offering the potential for more usable and responsive experiences [7, 1, 6]. While prior work has explored adaptation in

2D interfaces, specific application domains, and isolated use cases, there is no systematic review that brings together the recent surge of adaptive XR research. This paper fills that gap by reviewing the goals, strategies, and evaluation methods used in adaptive XR systems, identifying key trends and opportunities to guide future research.

Building on this, my PhD research investigates adaptive interaction in XR. Specifically, how interfaces can dynamically respond to users' physical, environmental, and cognitive conditions. Instead of treating all users the same, adaptive XR systems should adjust spatial layout, input strategies, feedback mechanisms, and interaction logic to meet the moment-to-moment needs of the user. Achieving this requires sensing, modeling, and responding to human factors in real time or near real time. My research addresses this challenge from following angles:

- I investigated how age-related differences affect interaction performance in XR. Using 3D target selection and manipulation task, I compared younger and older adults across different spatial regions in VR.
- I explore how supine posture affects interaction in XR. I design and evaluate a hand redirection technique that remaps interaction targets to more reachable areas while maintaining spatial coherence.
- I explore how adaptive MR learning environment that tunes task complexity and visual feedback in response to the user's cognitive load. This reflects a shift from physical to cognitive adaptation—critical in educational and training applications where mental engagement must be balanced with task difficulty.

## 2 PHD RESEARCH

### 2.1 Aging Effect in XR

Prior research shows that VR could be beneficial for older adults. For instance, it could potentially aid older adults in preserving or enhancing their physical and cognitive capabilities, encompassing aspects such as postural stability, reaction speed, and overall physical well-being. It could also promote their social and emotional well-being. Despite the potential benefits of VR, its interaction techniques have rarely been designed or studied with older adults. As people age, the changes in their perceptual, motor, and cognitive abilities can significantly impact their performance and experience when operating interactive devices. This consideration becomes particularly relevant when examining the aging effects on basic interaction tasks, such as object selection and manipulation, on 2D and 3D interfaces.

In order to gain an in-depth understanding of the user experience of older adults and the effect of aging on direct interactions (e.g., use handheld controllers to reach and contact virtual 3D objects directly) in VR between older and younger adults, we aimed to answer the following research questions (RQs):

- RQ1: How do spatial factors (e.g., target layout, distance from users, and interaction surfaces) affect the interaction performance and user experience of older adults when interacting with VR?

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- RQ2: How do their performance and user experience differ from younger adults?

In this study, we aimed to evaluate the performance and accessibility of different spatial factors (i.e., position) in VR space. Specifically, we evaluated and compared the object selection and object manipulation performance and perceived difficulty between older and younger adults.

The results highlight clear and systematic differences between older and younger adults. Older adults exhibited longer movement times and reduced accuracy, particularly for targets placed in extreme vertical or lateral directions, where limited neck mobility and reduced shoulder range caused substantial increases in effort. They were more likely to over- or under-shoot targets during the correction phase, indicating reduced fine-motor control and slower sensorimotor integration. Participants also reported higher physical fatigue, especially in the neck and upper arms, when interacting with out-of-view or hard-to-reach elements. Together, these findings show that aging introduces distinct spatial asymmetries and ergonomic vulnerabilities in VR interaction, underscoring the need for adaptive systems that adjust interface placement and interaction demands according to user ability and spatial context.

## 2.2 Ongoing-Work

### 2.2.1 Posture Effect in XR

Most VR interaction techniques are designed for upright postures, such as sitting and standing. While lying-down VR is effective for passive activities like watching videos or meditation, interactive tasks such as gaming or 3D object manipulation remain challenging. Unlike upright postures, the supine position limits reachability and makes distant interactions more difficult, as users cannot lean forward to extend their reach as they can in upright postures.

To address this limitation, reducing hand movement distance is essential, previous research has explored hand-remapping techniques that amplify virtual hand movements, reducing the need for extensive physical motion. However, these methods are designed for upright postures, assuming a user's natural reachability and control space without considering the constraints imposed by lying down. This project will model the comfort region and design an interaction technique in supine posture.

### 2.2.2 Adaptive Cognitive Load Modeling

Embodied learning leverages gesture, movement, and spatial interaction to deepen conceptual understanding, and Mixed Reality (MR) provides an ideal environment for such learning by blending physical action with immersive digital content. However, this richness also introduces substantial cognitive demands: learners must process sensory information, interpret spatial layouts, and manage complex interfaces while engaging with abstract concepts. According to Cognitive Load Theory, excessive intrinsic or extraneous load can overwhelm working memory and diminish learning effectiveness—an issue amplified in MR, where embodied actions and interface navigation compete for attention. Existing adaptive XR systems primarily adjust interface layouts but rarely address deeper cognitive demands. To unlock the full potential of embodied learning, MR environments must intelligently regulate cognitive load in real time. This project will develop an Adaptive Cognitive Load Regulation Model for MR learning.

## 3 CONCLUSION

In summary, my doctoral research aims to advance adaptive XR systems that intelligently personalize interaction, learning, and accessibility through integrated sensing, reasoning, and adaptive interface design. As wearable sensors, large language model technologies continue to evolve, they offer unprecedented opportunities for XR systems to become more context-aware and human-centered. Understanding how these technologies can be leveraged

to support diverse users and usage scenarios remains an exciting and open challenge. Moving forward, I hope to refine the theoretical foundations, methodological approaches, and practical design strategies needed to build robust adaptive XR systems.

To guide this next stage of research, I would value feedback from the Doctoral Consortium on the following questions:

- How can wearable devices be effectively integrated into XR to support continuous sensing and context-aware display without compromising comfort or usability?
- What role should intelligent agents play in adaptive XR reasoning, and how can we ensure their decisions remain transparent, reliable, and aligned with user intent?
- How can dynamically generated or procedural XR user interfaces be constrained to ensure ergonomic safety, cognitive clarity, and accessibility for diverse users?

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